



Point Estimation Transcript

Now, we are going to know how do we estimate about the entire population using samples. So, there are two different types of things that we do, one point estimate, the other one is an interval estimate and in this section we are going to learn about a point estimate. So, point estimate is all about making the best guess and the best guess may sometimes be the wrong guess as well.

So, we are going to learn about when not to be wrong. So, let us say for example you have a jar full of toffees or gumballs or whatever and you would want to estimate the average weight of the entire jar of toffees. Now you might already know how much a toffee weigh or you might already know about different other statistics, but if you have to weigh the entire jar or the weight of the entire toffees full of jar, what you can do is you can take some sample or let us say scoop of toffees and then weigh them and use that statistics to understand how much could be the entire weight of the toffee full of jars, if you know how many toffees are there.

So, using all of that you can easily very well easily estimate about what exactly is the weight. Similarly, let us say if you would want to know about the proportion of people who have voted for Congress or BJP in 2019 election, you might want to go ahead and take sample from various different regions in the country and then ask them about whom they have voted and then reduce that percentage by taking the ratio of total number of people which would be in the denominator and then in the numerator which party they voted from. So that would be a proportion.

Similarly, you would estimate that then you would say that let us say that the output comes out to be 50% of the people voted for Congress and 50% of the people voted for BJP. Now that output you would say is very likely associated with the population. So there are these statistics which are single point estimate proportion because that's a single numbers.

Let's say for example a very simpler analogy that I can let you understand with is you meet an individual and you would want to estimate the height or let's say the age of the individual and to estimate the age you would look towards their features as in the colour of ears, whether they have beards or not, whether they have wrinkles on the faces or not and then you estimate age of let's say 20 years but that 20 might not be the right value. So that 20 might not be the right value for the individual, the individual might be older than that or the individual might be younger than that. So how would you decide the point estimate or the single number, the single good estimate that you have devised or that you have taken or that has come into your mind or that you have found using some method, whether it is right or wrong, whether it is actually very true or it is very closer to the true value, true value as in the actual value of the individual's age right you are investigating.



So all of these things is what we are going to learn in this section and we will first get introduced to what different types of point estimates are. So there are three different types of point estimates that we generally use. I mean when you have the definition of point estimate as a single number that could be anything but generally in statistics we deal with these three things, one is called as population mean, the other one is called as population proportion, the third one is called as population variance.

If you think about knowing about the population parameter which is population mean, population proportion, population variance. So think of population mean as the height example or the age example that I gave you right. Population proportion could be the party voted for, how many individuals or how much or what proportion of individuals has voted for Congress or BJP right.

So let us say 10% have voted for BJP or 20% or 80% have voted for BJP that is a single statistic. Similarly population variance, population variance is let us say is the variance 3, 10, 300, 3000 what, what variance mean is how much is the spread whether that spread is 2, 3, 4 but that variance is about the spread of the data but use to understand what is the spread we have a single number. So variance is also a point estimate.

Similarly these are parameters population mean, population proportion, population variance all of these are point estimate. To know about it we take samples. So those sample statistics are called as sample mean, sample proportion and sample variance.

Now there are some properties of the sample statistic that we are taking let us say sample mean or sample proportion or sample variance or any other statistic that could come in our mind. The two properties should very well be holding true for them. The first one out of these is they should be unbiased.

Now look at the figure that I have on the left hand side I have an unbiased estimator basically because there is a lot of variance in those blue points. If you think of, if you are trying to estimate average height of people living in India and this yellow region is the true value which might be 5.5. So your sample one of the sample has let's say 5.6 the other sample has 5.7 where my cursor is and the other sample has 4, 3 and then if you average them all together you might reach to a true value which is 5.5 or you might reach to yellow right. So that is unbiased because you are still able to though your observations are very very varied but you are still able to get to the true value.

Here in on the right hand side the biased estimator we have a lot of those observations which are far far away from the target right. Now because they are far away from the target even if you take a proportion or average of them it would always be far away from the true value which is the yellow part right. So biasness is you are systematically away from what the actual value or the true value is and in the right hand side figure if you see that is what is happening.



So the very first property is the estimator has to be unbiased. Second thing is the estimator also has to be very much consistent. Now on the left hand side you have a more efficient estimate because you see all of those viewpoints they are exactly on the target right.

There is no variance at all. So they are more consistent in terms of the type of observation that you have taken while on the right hand side you have more variance they are less efficient. So there are two different types of things that we take into consideration.

The very first thing is unbiasedness the second thing is efficient they have to be unbiased and they have to be consistent which means efficient right. Now let us say this is an interactive example that we created let us say you would want to devise two different estimators and both of these are point estimates. The first point estimate on your screen is sample mean.

The second point estimate is a sample mid range which is the average of minimum and maximum or let us say some of them and divided by 2. So that is a midpoint of your minimum and maximum and the other good estimate is sample average. Now let us say you take a simple sample of 30 people and once you take a sample you have green line which is basically the mean and you have orange line which is the second estimator which is the midpoint of min and max and red one is the true value. Now once you have taken the first sample or let us say once you take the second sample third sample you would generally see the green line would be more closer to true value and that is what I meant when I said the estimate has to be unbiased it means that it might vary but it has to be closer to the true value right.

The second thing is it needs to be consistent see with every sample that I am taking there is less variance in the green line but there is more variance in the orange line. When I am clicking draw one sample you could see that orange line is more getting here and there while the green line is not right. So that is what it means.

So these are the two different properties I hope this was understood to all of you. Now one of the easiest understanding also we have to relate to unbiasedness is you all might have seen the formula for variance. The formula for variance has $n-1$ in its denominator.

Do you know why that is? Let me help you understand why exactly the formula for variance has $n-1$ in it. So let us say the formula for variance is variance is equals to summation of each observation minus its mean how far the individual observations are from the average value divided by $n-1$. So basically at the top you have the deviation square and at the bottom you have $n-1$.



Why is there $n-1$? Why not n ? This can be very well understood using a very simple analogy then I will relate that analogy to this example. So let us say you have a football field and in that football field you have to choose 10 players and you are playing a strategy wherein you have 4 forwards, 3 mids and 3 people at the back. Now let us say the very first individual the very first player arrives and they have the degrees of freedom to choose any position or they have the freedom to choose any position that individual can very well go here or they can go here and they can choose any of these positions.

Now similarly let us say the first individual has taken this position. Let us circulate with a different colour. So this position is already taken.

Now think of the second individual the second player comes in and they decide to choose another position which is this. Now in this process if all the 10th player comes in and they choose randomly different different types of positions which are vacant, 9 of them could choose all of these positions. Now think of the 10th individual all the individuals let us assume that all the individuals have taken the position.

Think of the 10th individual, 10th individual has only one place left which is this one. Now 10th player does not have any freedom to choose its position and that is why 10th player cannot vary its position. Similarly when you are estimating variance from a sample you are already taking mean from the sample and then you are saying how does one individual observation varies from the mean.

Now in this entire process if you think of mean itself to come up to a particular mean if you are taking mean of 10 different values or observations let us say x_1 plus x_2 plus and so on plus x_{10} divided by 10. So if you think of in this entire formulation to get to a particular value of $\bar{x}$ which might be let us say a constant value of 5.5 let us say it is a height to get to 5.5 all but not one could vary. So let us say all of them varies but one does not have the flexibility to vary and that is why we have this formulation of $n-1$ which relates the concept of degrees of freedom or how many entities are free to revolve or oscillate or vary and so on.

So that is why we have this basal correction which is the formula for variance which has been written like this. Now in this entire section we have discussed about various different types of things we went and discussed about point estimates what exactly is an estimator properties of estimator right estimator could be three different types or many different types of point estimator and general as could be many different types but we studied three different types of specific estimators one point estimator as mean second as proportion third as variance. These majorly cover majority of the things that we will be learning.



With that I will conclude this section and then we will move to confidence interval.
Thank you so much.

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